Dempsey Center Report

Lulu Xu, Honglin Chen

Introduction

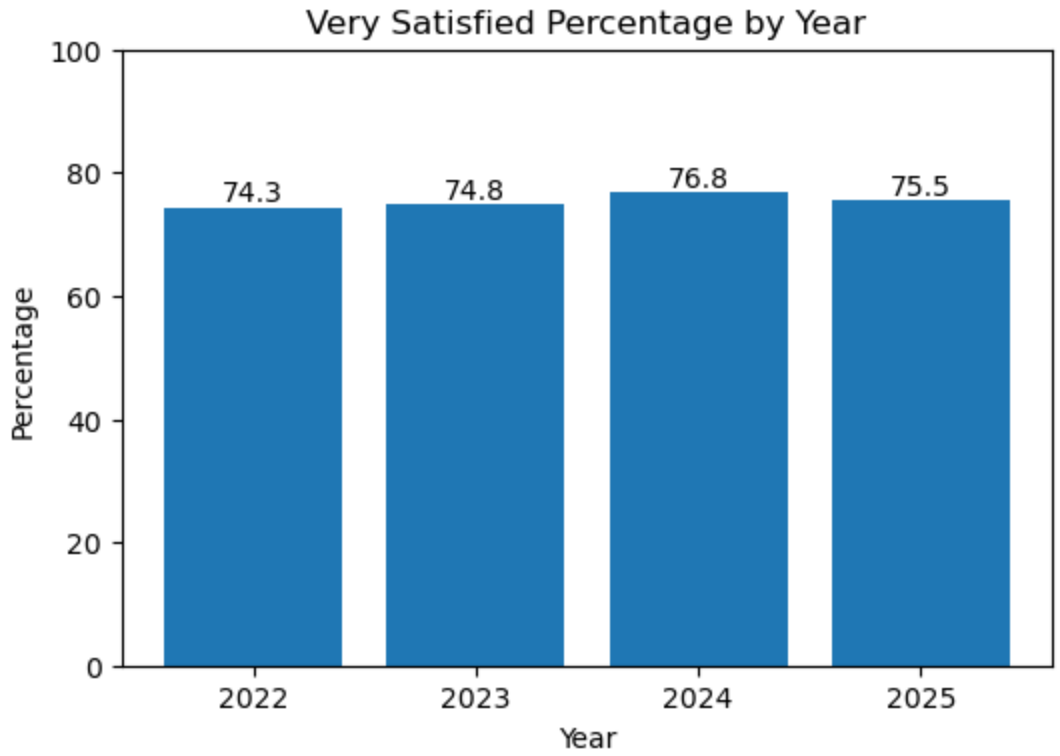
- This project analyzes the Dempsey Center Annual Client Survey data from 2022 to 2025
- The goal is to understand client satisfaction, perceived impact, and likelihood to recommend
- We first perform exploratory data analysis (EDA), followed by deeper analysis

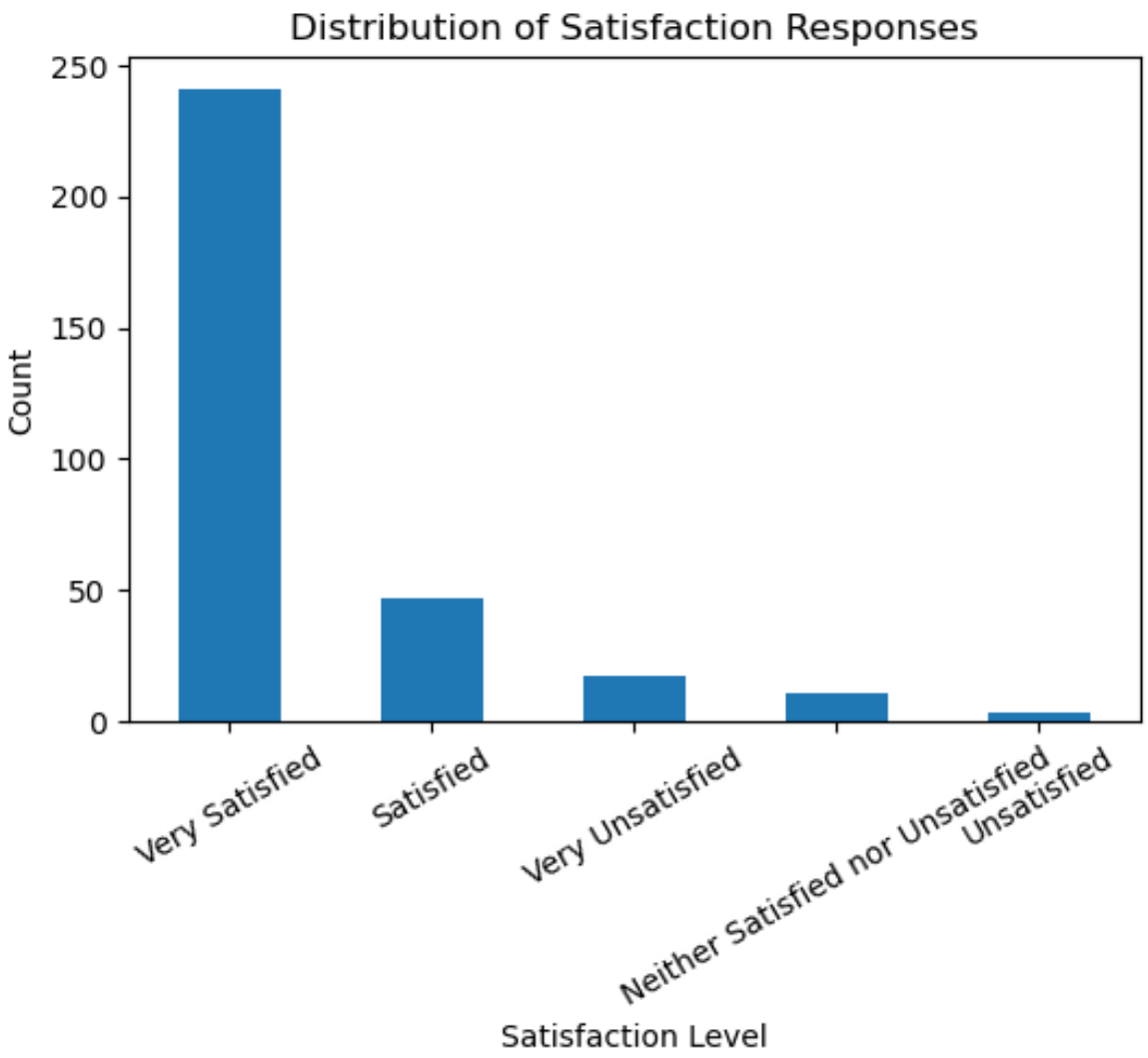
Overview of Key Metrics

- Satisfaction (Very Satisfied ~75%)
- Impact (Made life better ~95%)
- Recommendation (~9.6/10)

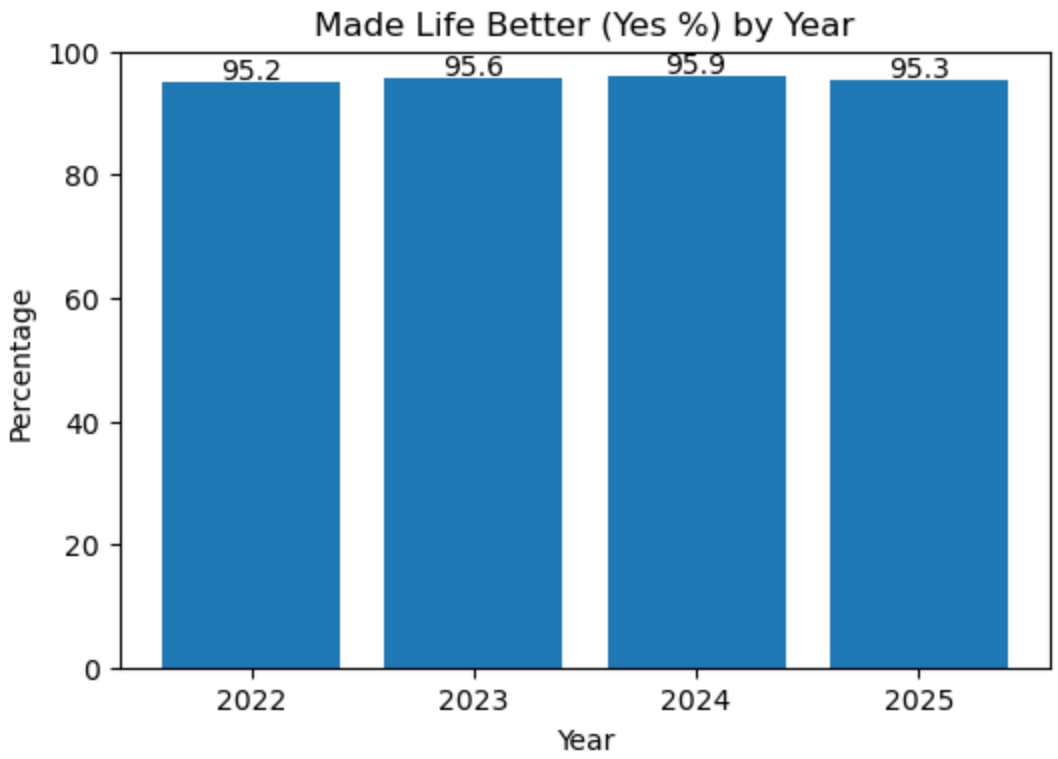
Basic Summary

Overall Satisfaction

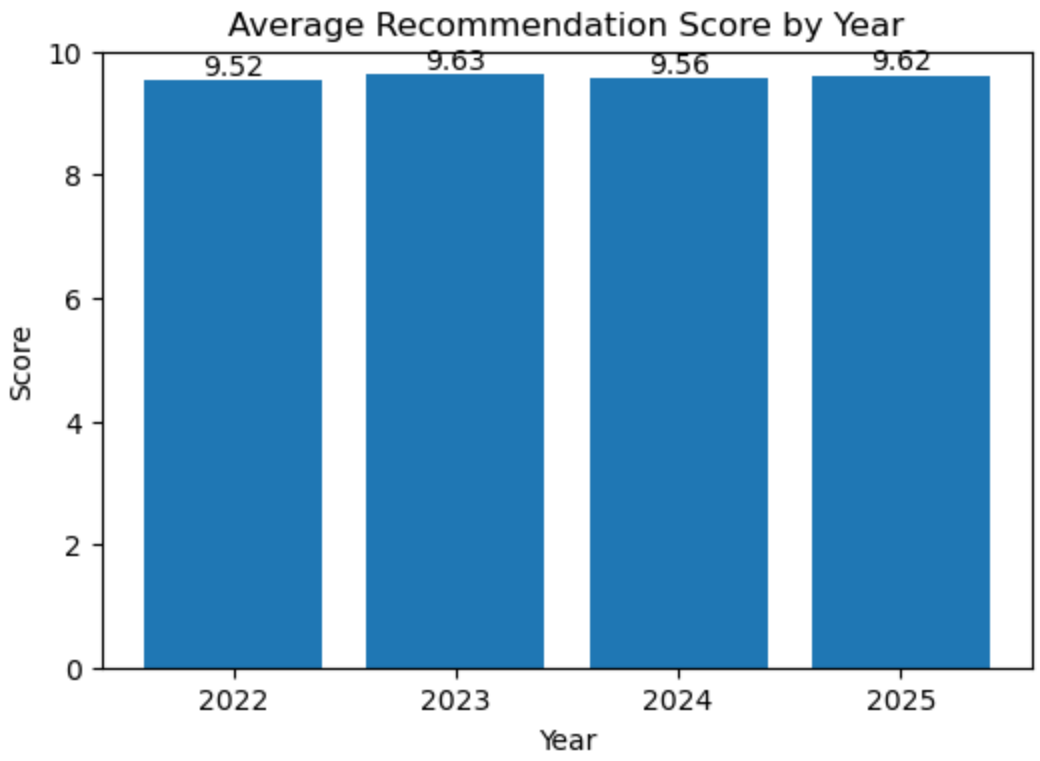




Perceived Impact of Services



Likelihood to Recommend

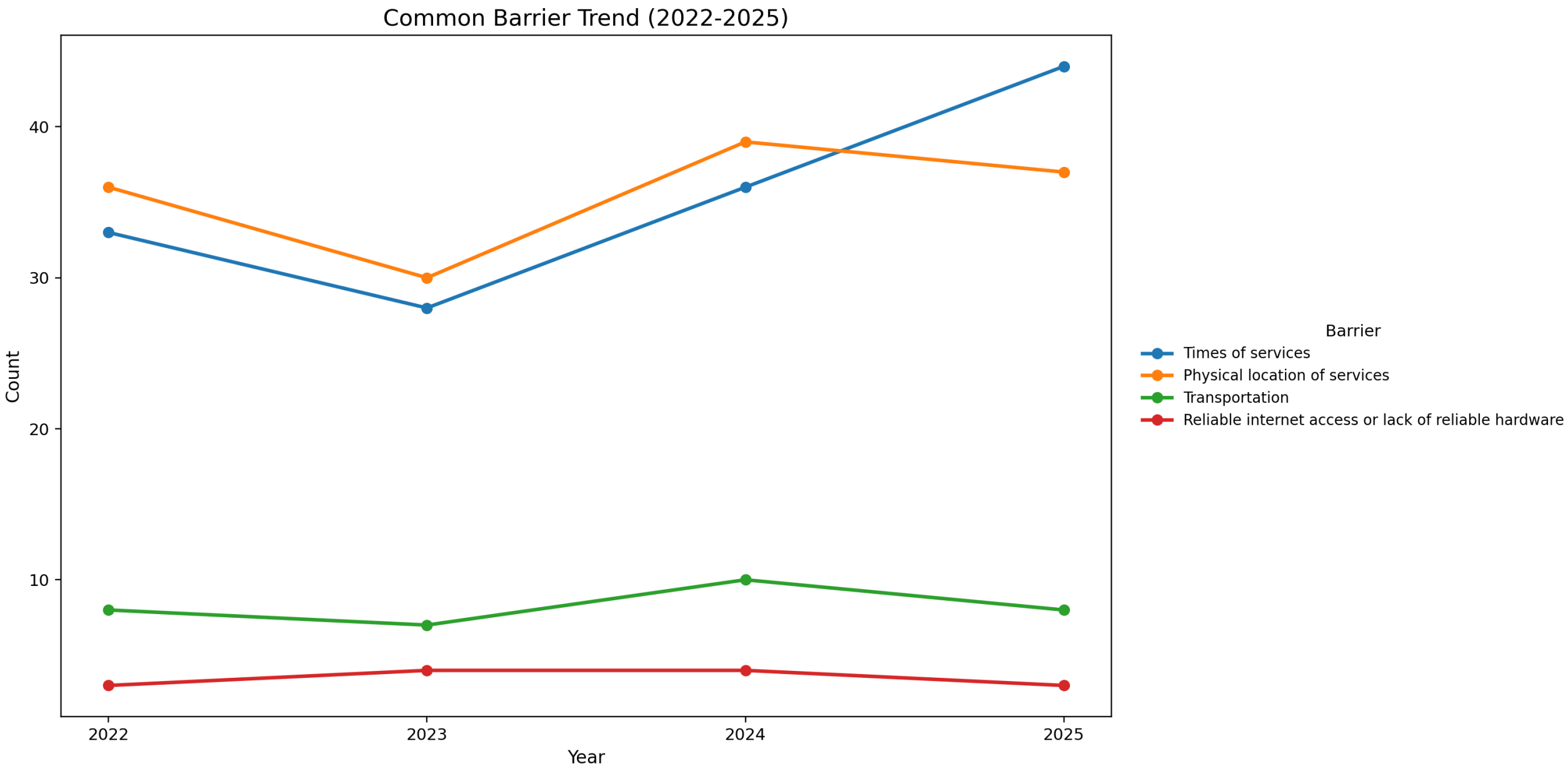


Key Takeaways

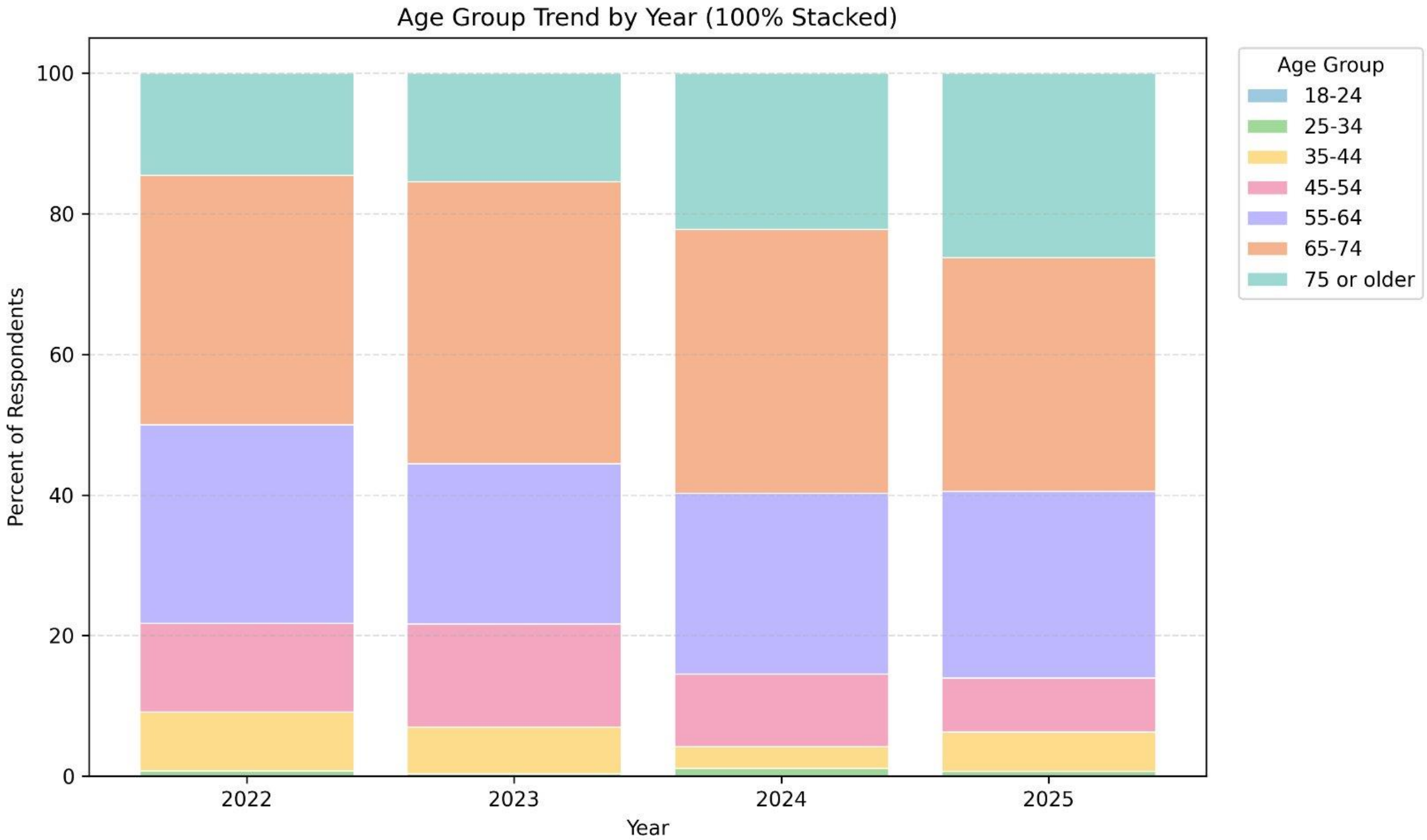
- Satisfaction is high and stable
- Perceived impact is extremely strong (~95%)
- Clients are highly likely to recommend (~9.6 out of 10)

Analysis – Barriers, Age, Services

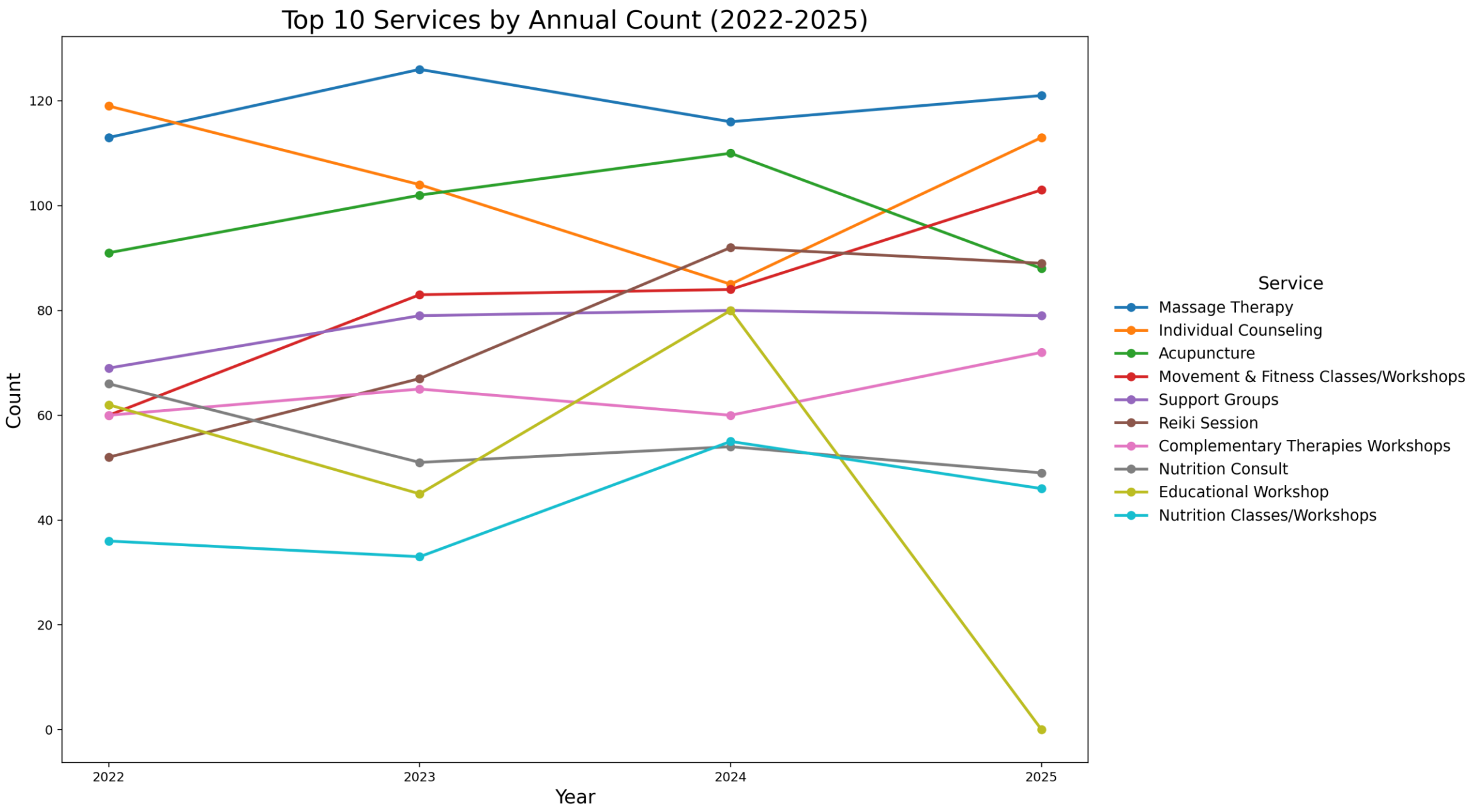
Barriers

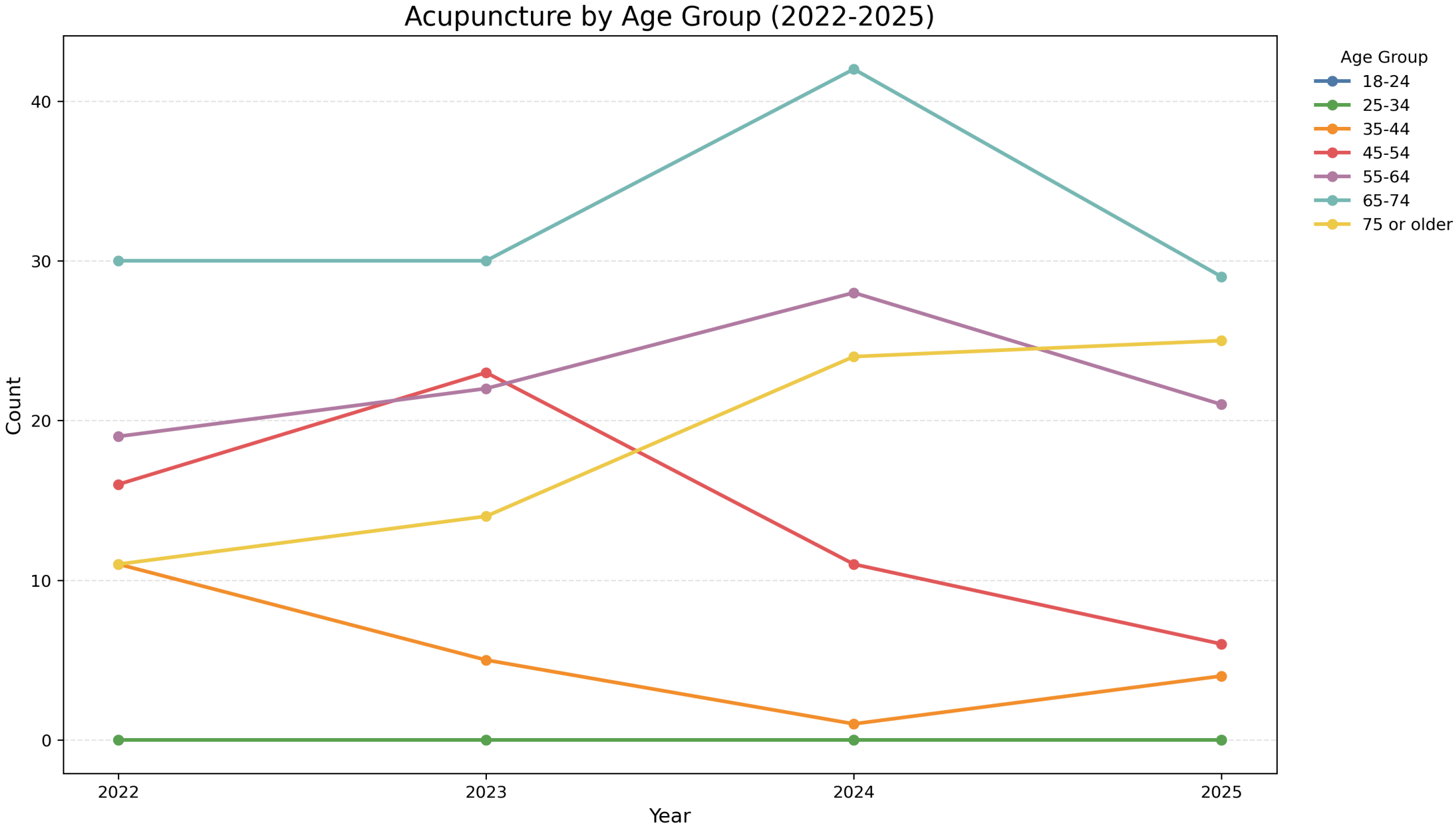

Common Barrier Trend (2022-2025)
Count
40
30
20
10
2022
2023
2024
2025
Year
Barrier
Times of services
Physical location of services
Transportation
Reliable internet access or lack of reliable hardware

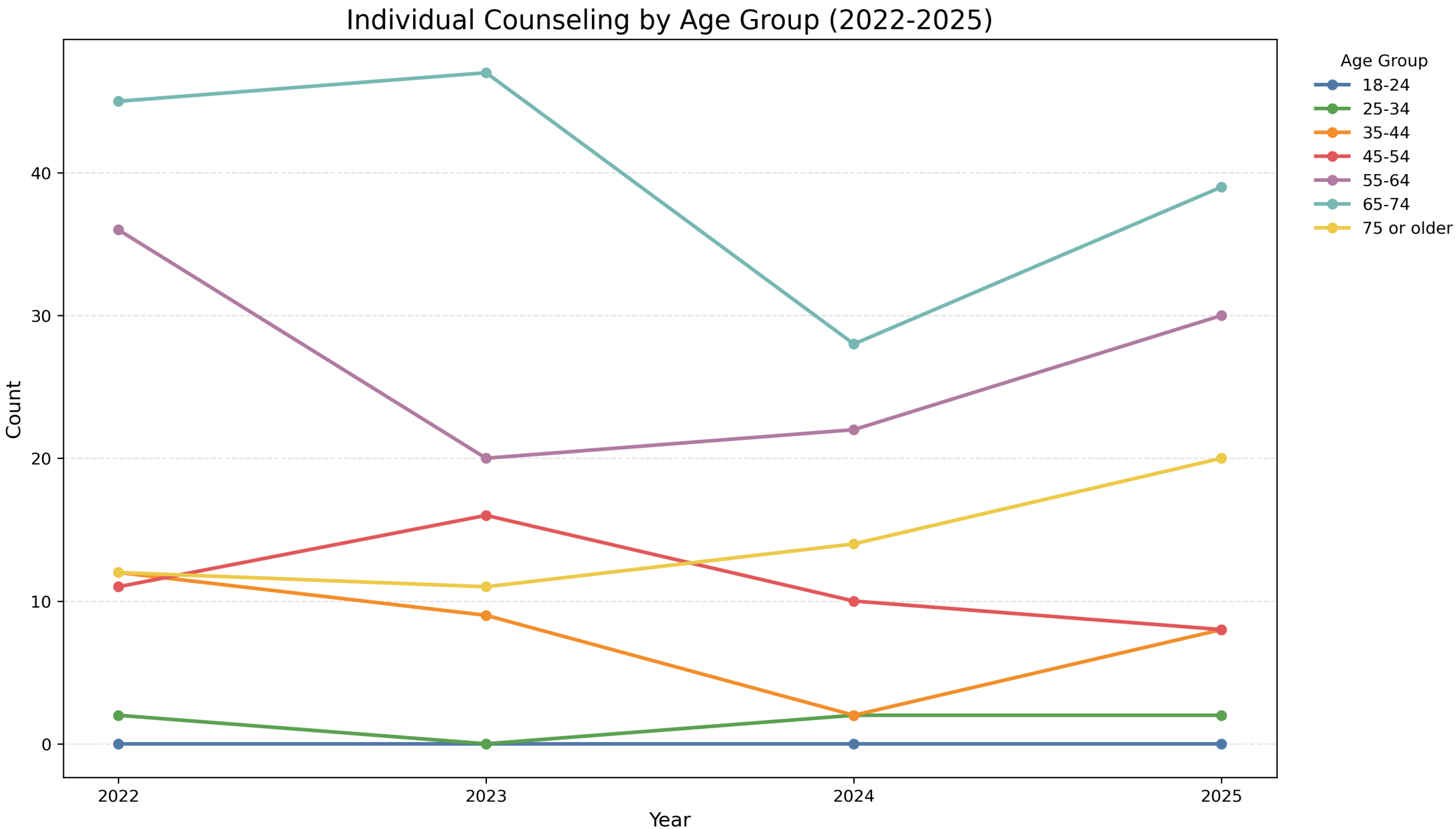
Age

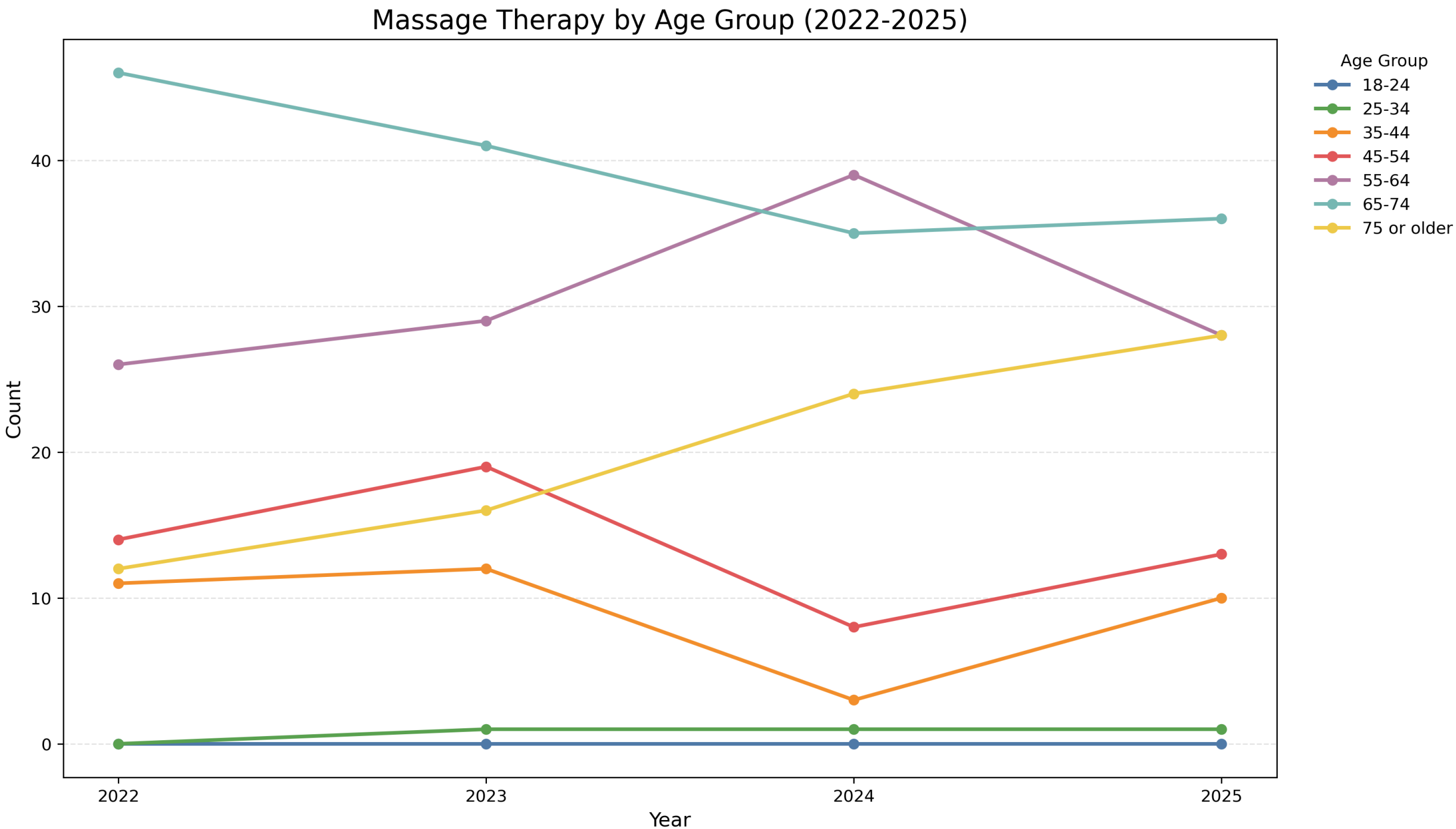

Age Group Trend by Year (100% Stacked)
Percent of Respondents
100
80
60
40
20
0
2022
2023
2024
2025
Year
Age Group
18-24
25-34
35-44
45-54
55-64
65-74
75 or older

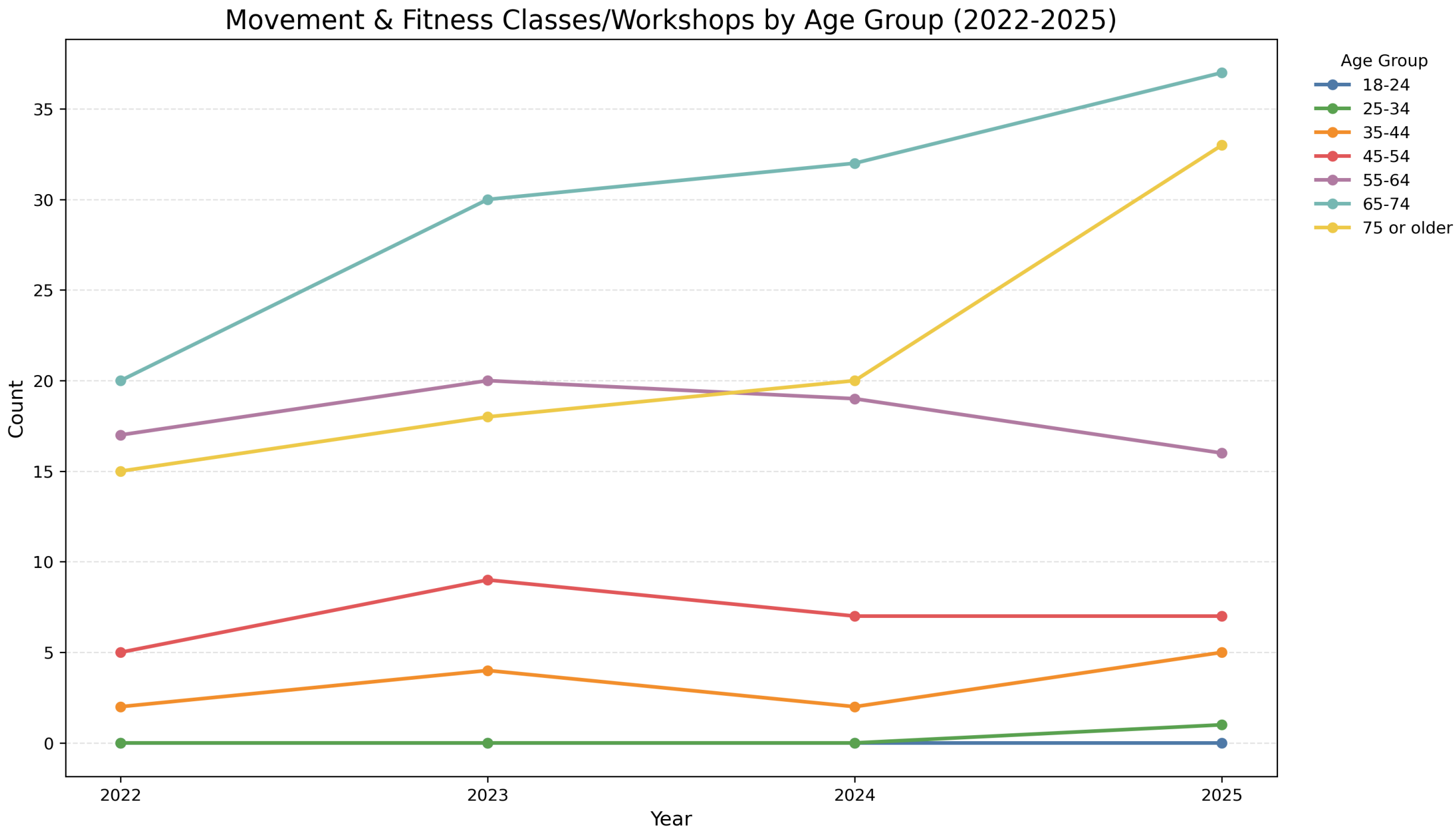
Services

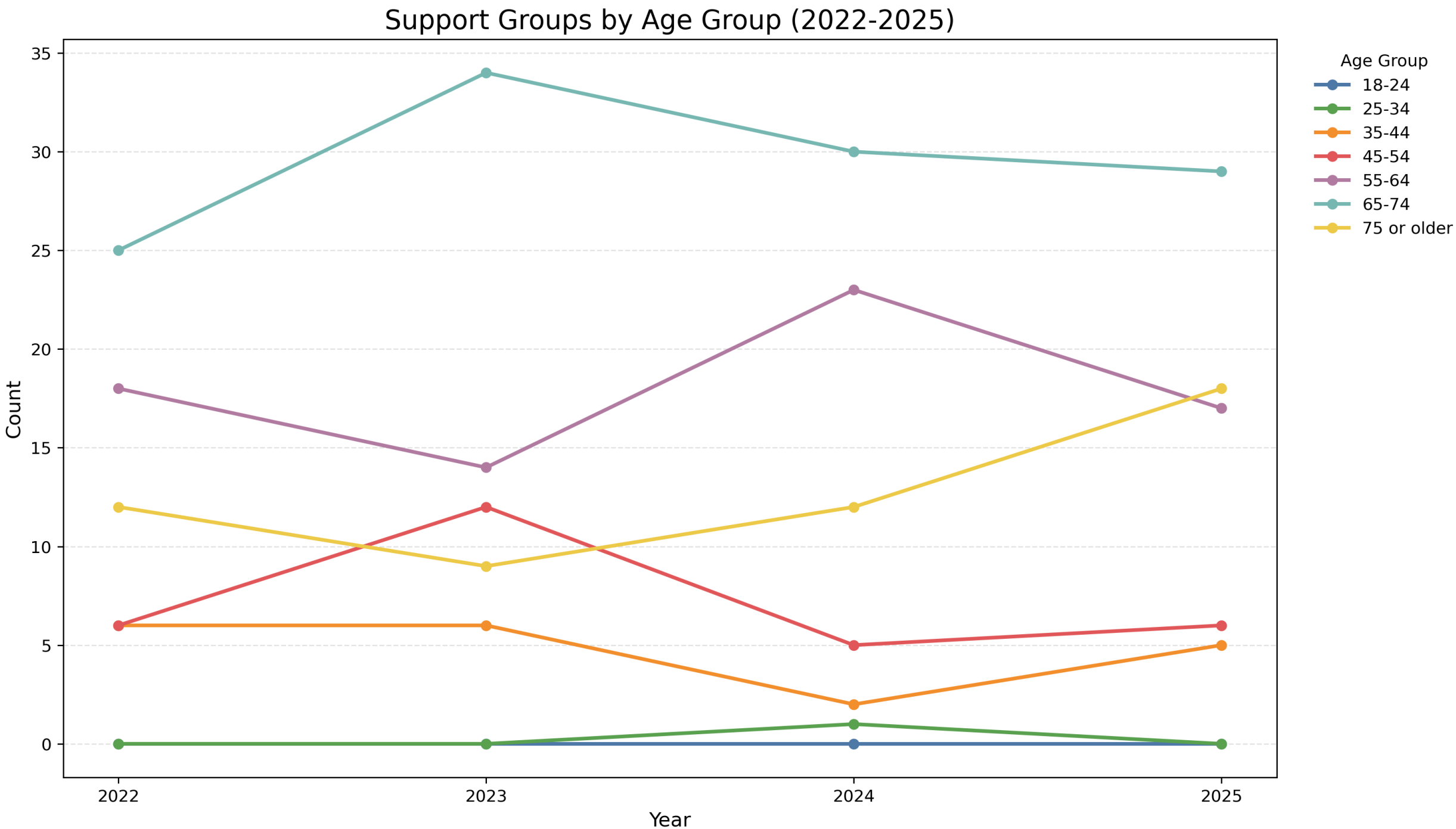

Top 10 Services by Annual Count (2022-2025)
Count
Year
0
20
40
60
80
100
120
2022
2023
2024
2025
Service
Massage Therapy
Individual Counseling
Acupuncture
Movement & Fitness Classes/Workshops
Support Groups
Reiki Session
Complementary Therapies Workshops
Nutrition Consult
Educational Workshop
Nutrition Classes/Workshops


Acupuncture by Age Group (2022-2025)
Count
Year
0
10
20
30
40
2022
2023
2024
2025
Age Group
18-24
25-34
35-44
45-54
55-64
65-74
75 or older


Individual Counseling by Age Group (2022-2025)
Count
Year
0
10
20
30
40
2022
2023
2024
2025
Age Group
18-24
25-34
35-44
45-54
55-64
65-74
75 or older

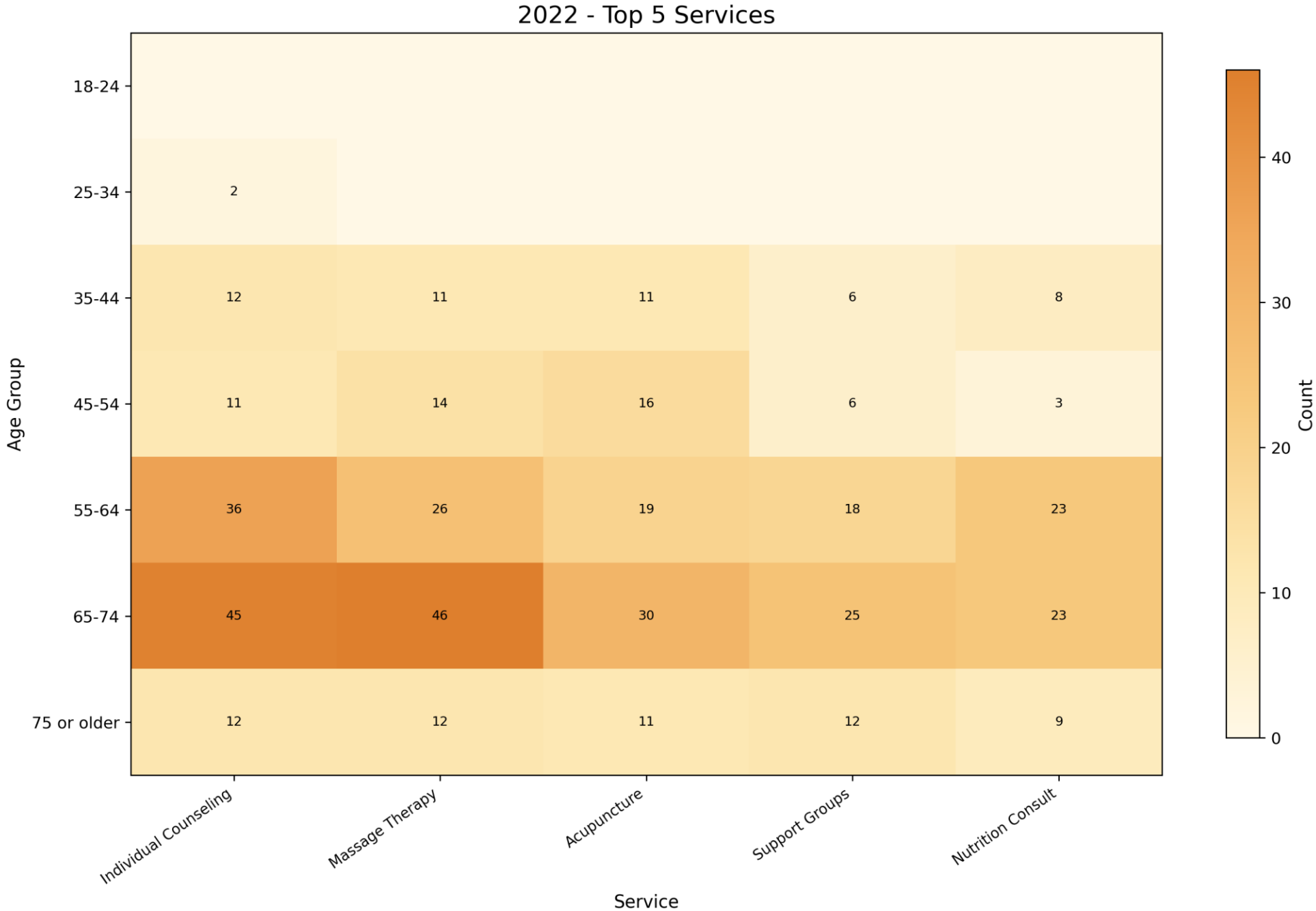

Massage Therapy by Age Group (2022-2025)
Count
Year
0
10
20
30
40
2022
2023
2024
2025
Age Group
18-24
25-34
35-44
45-54
55-64
65-74
75 or older


Movement & Fitness Classes/Workshops by Age Group (2022-2025)
Count
Year
0
5
10
15
20
25
30
35
2022
2023
2024
2025
Age Group
18-24
25-34
35-44
45-54
55-64
65-74
75 or older


Support Groups by Age Group (2022-2025)
Count
Year
35
30
25
20
15
10
5
0
2022
2023
2024
2025
Age Group
18-24
25-34
35-44
45-54
55-64
65-74
75 or older

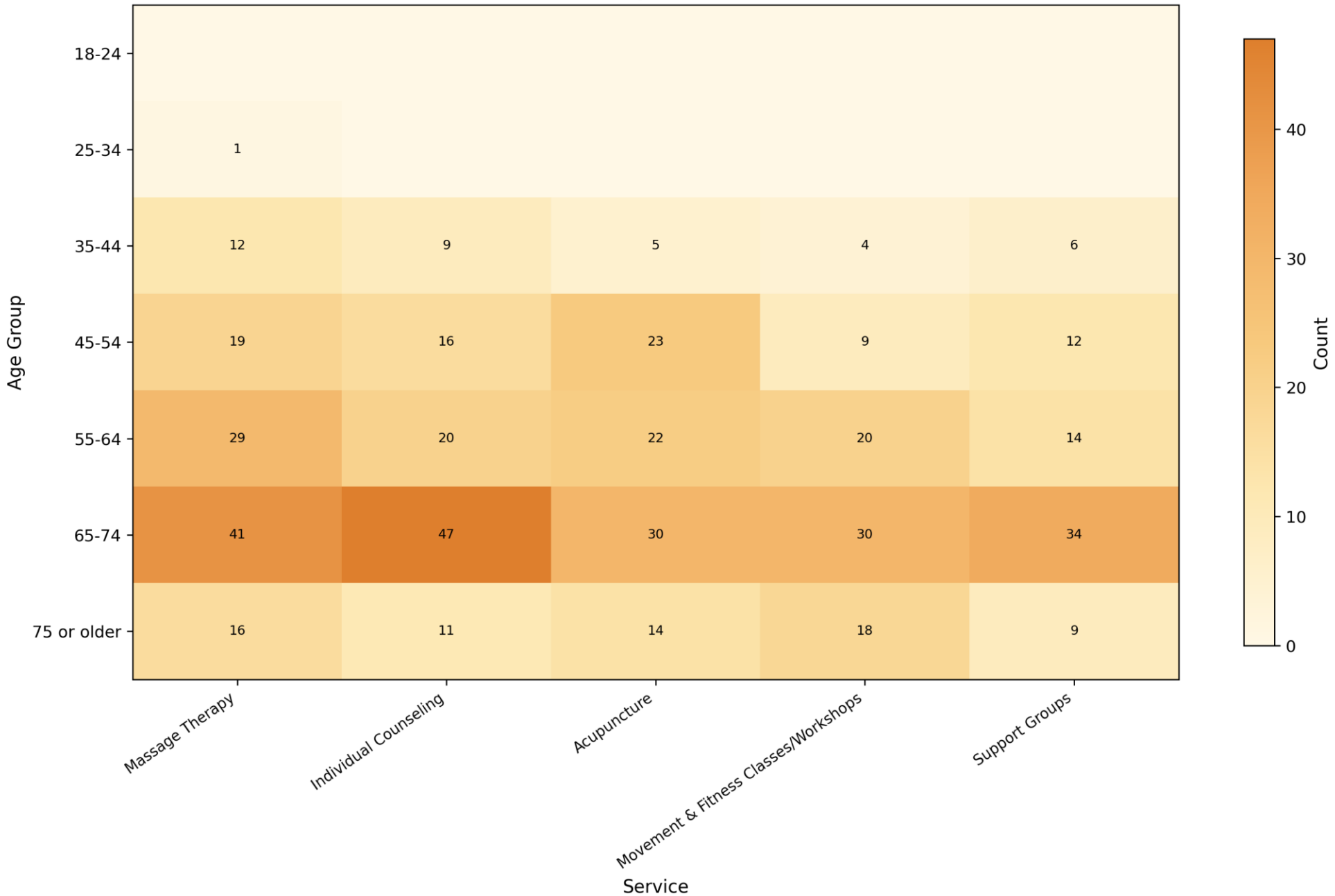
Services vs. Age

Age vs Services Heatmap - 2022 (Top 5 Services, Counts)



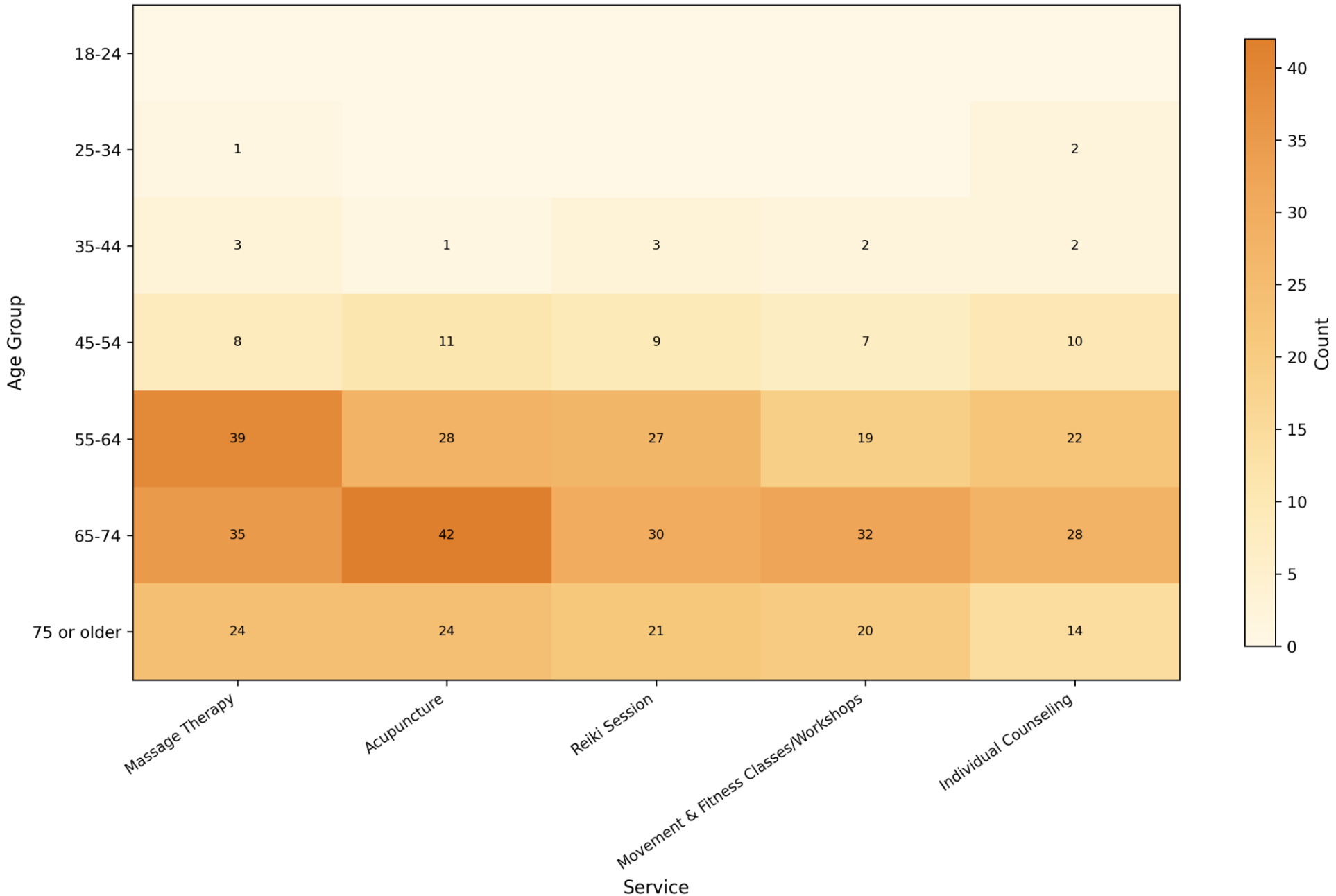
Age vs Services Heatmap - 2023 (Top 5 Services, Counts)

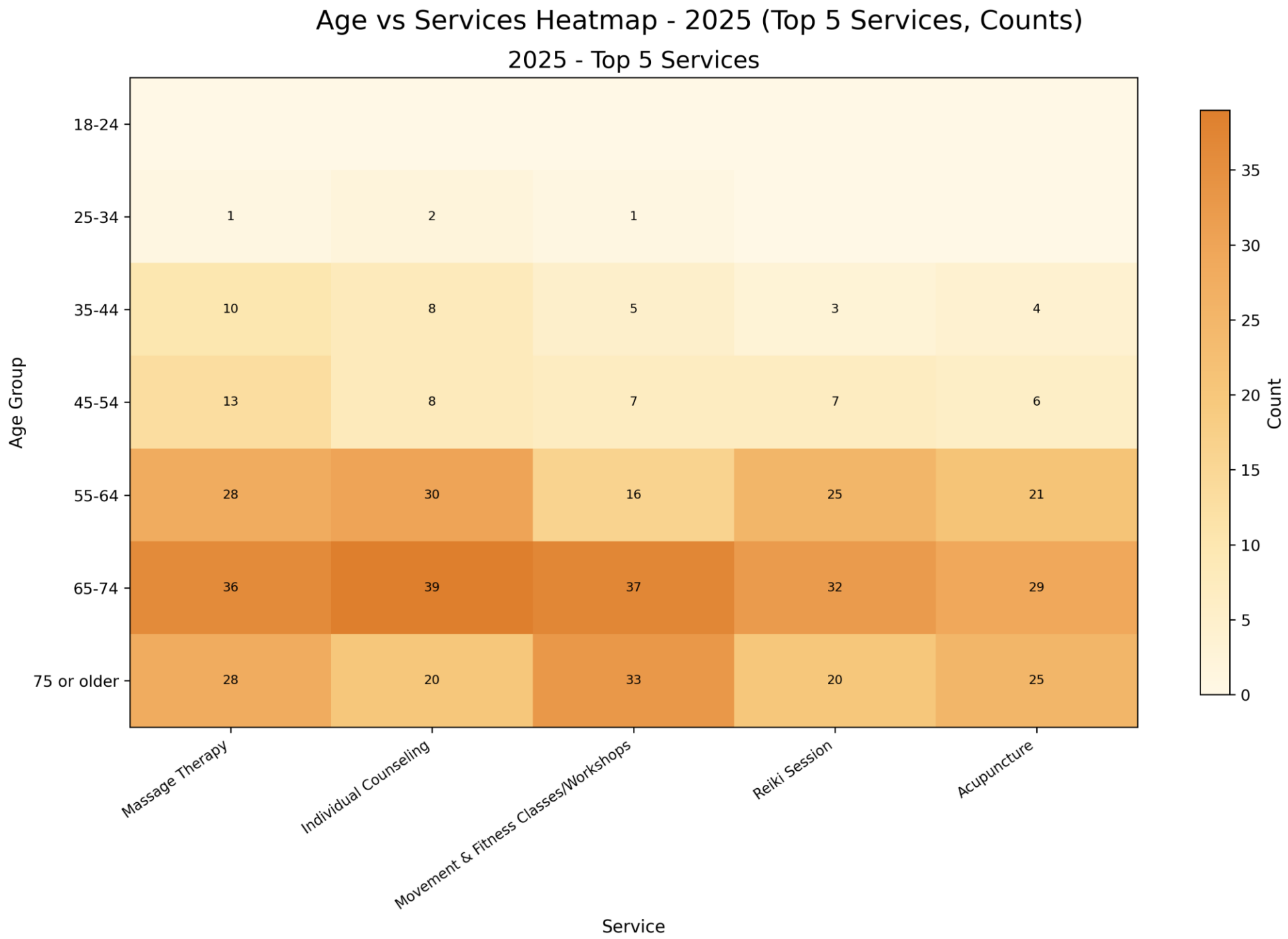
2023 - Top 5 Services



Age vs Services Heatmap - 2024 (Top 5 Services, Counts)

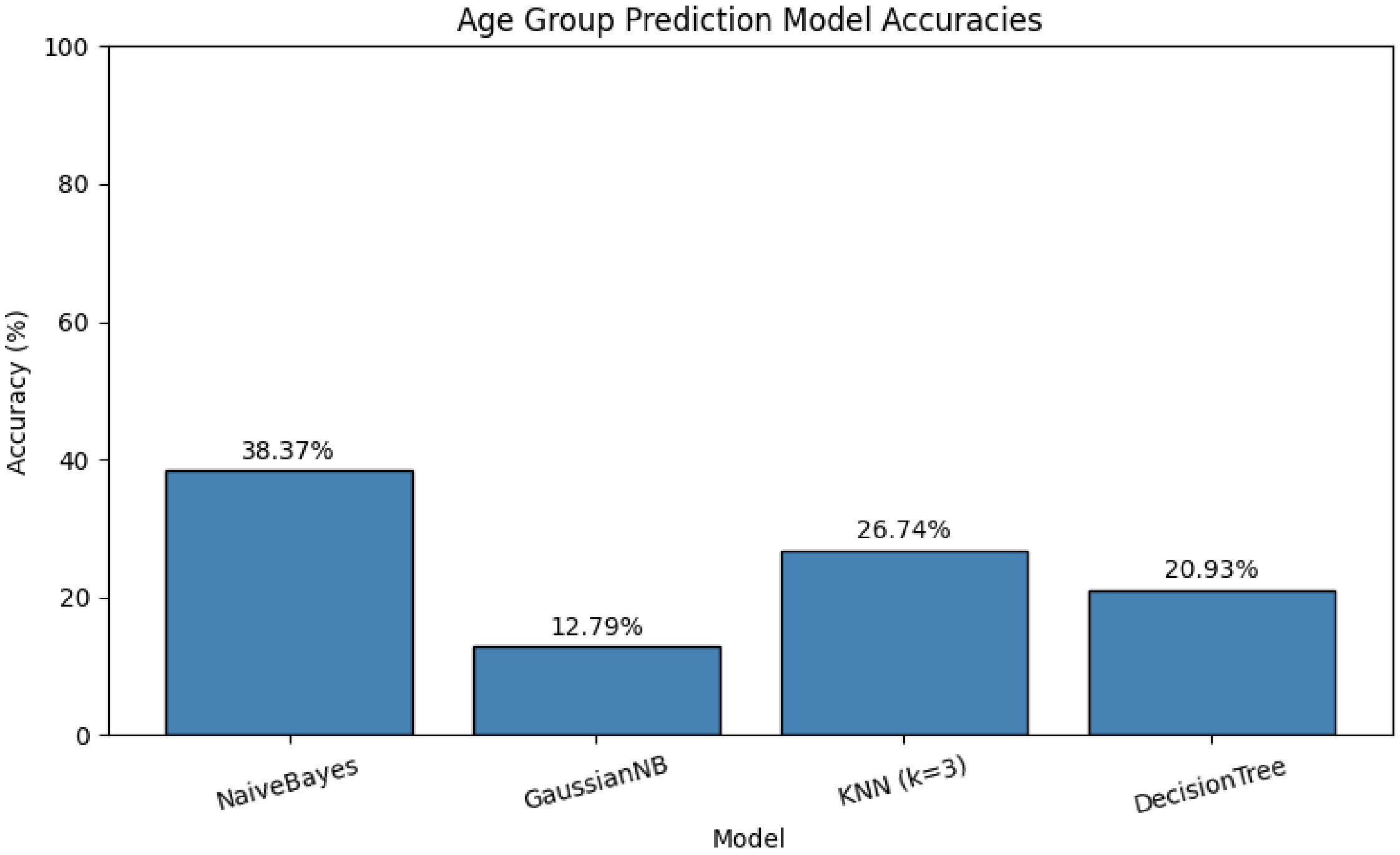
2024 - Top 5 Services

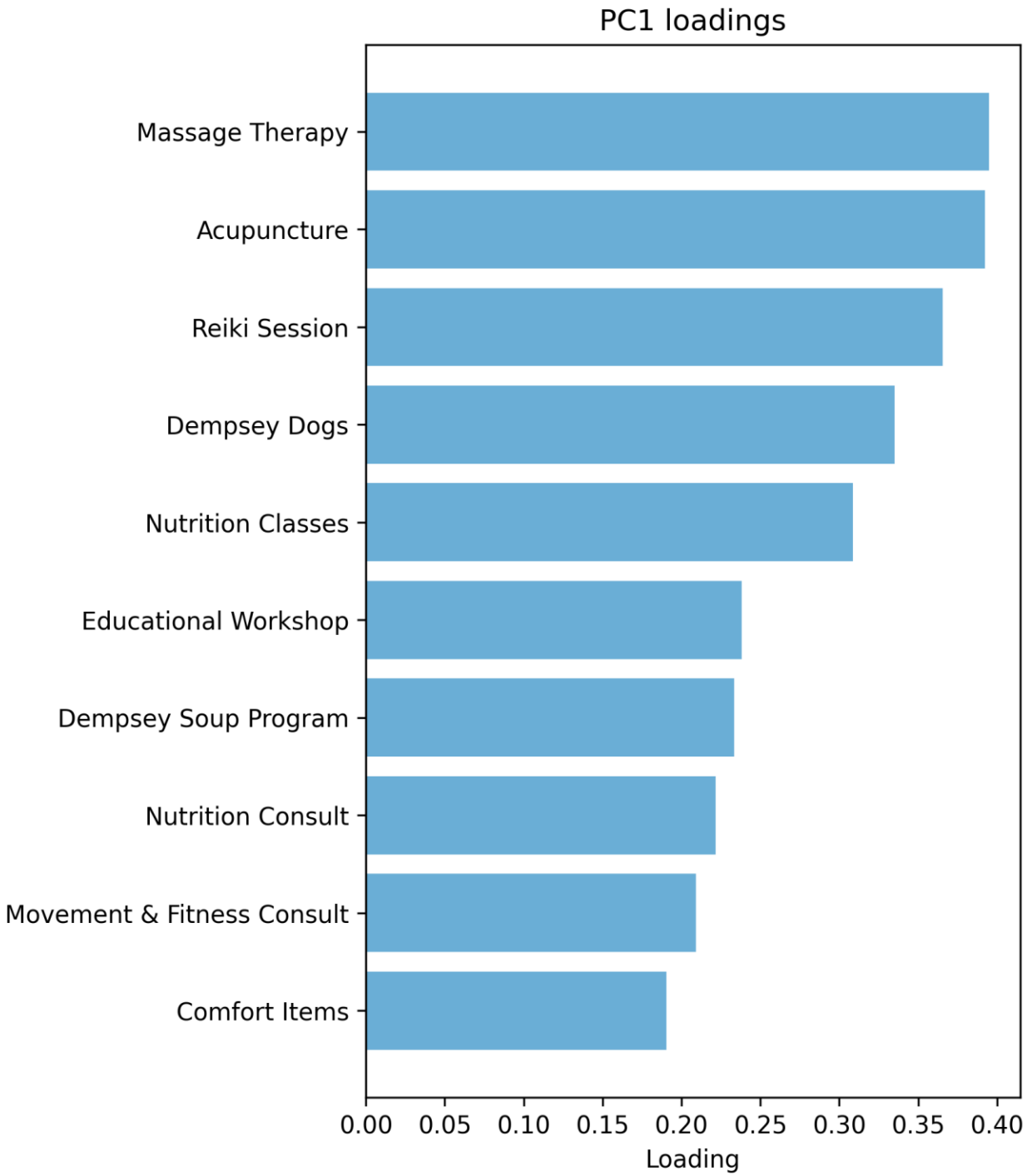


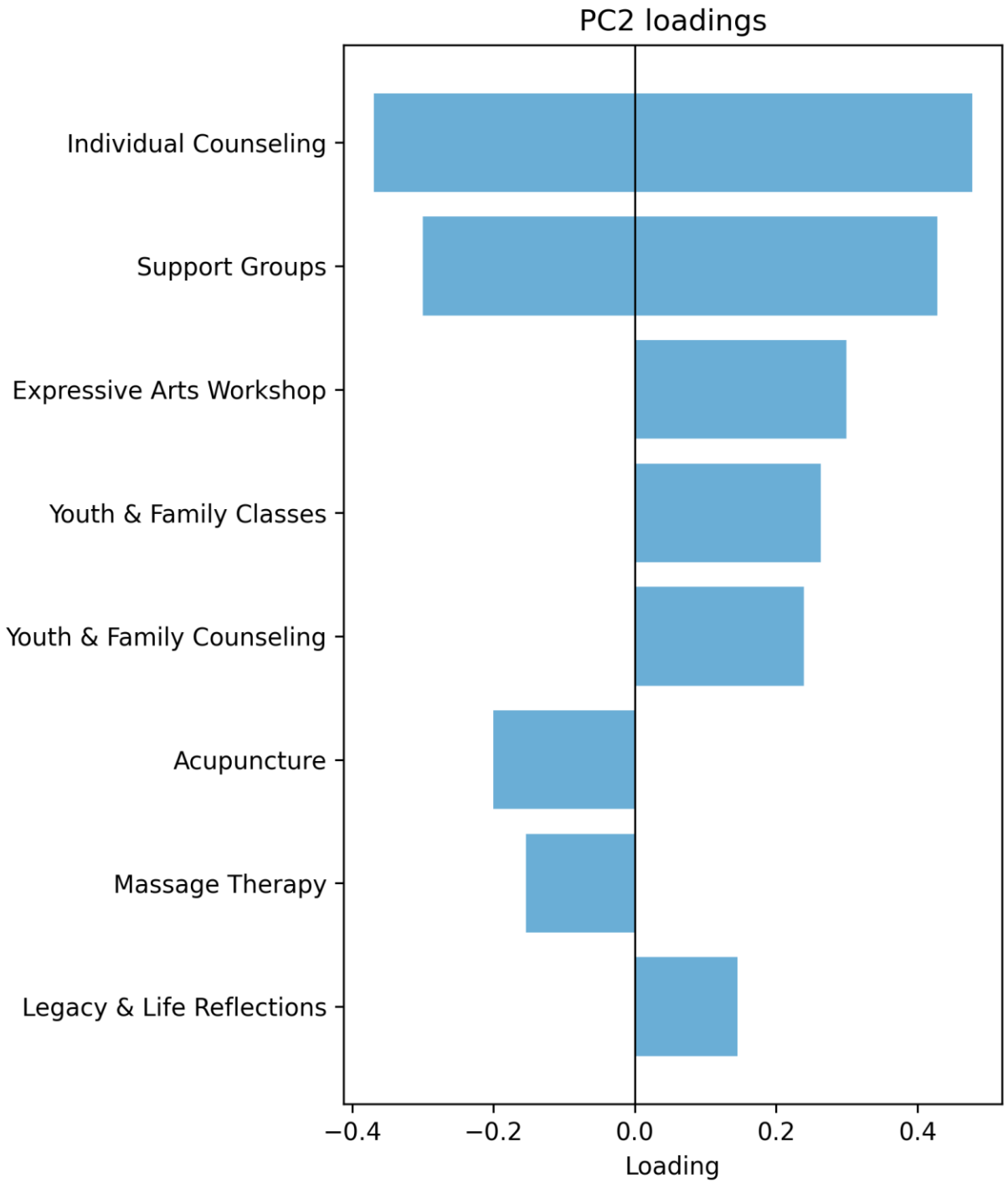

Age vs Services Heatmap - 2025 (Top 5 Services, Counts)
2025 - Top 5 Services
Age Group
18-24
25-34
35-44
45-54
55-64
65-74
75 or older
1
2
1
10
8
5
3
4
13
8
7
7
6
28
30
16
25
21
36
39
37
32
29
28
20
33
20
25
Massage Therapy
Individual Counseling
Movement & Fitness Classes/Workshops
Reiki Session
Acupuncture
Service
Count
0
5
10
15
20
25
30
35

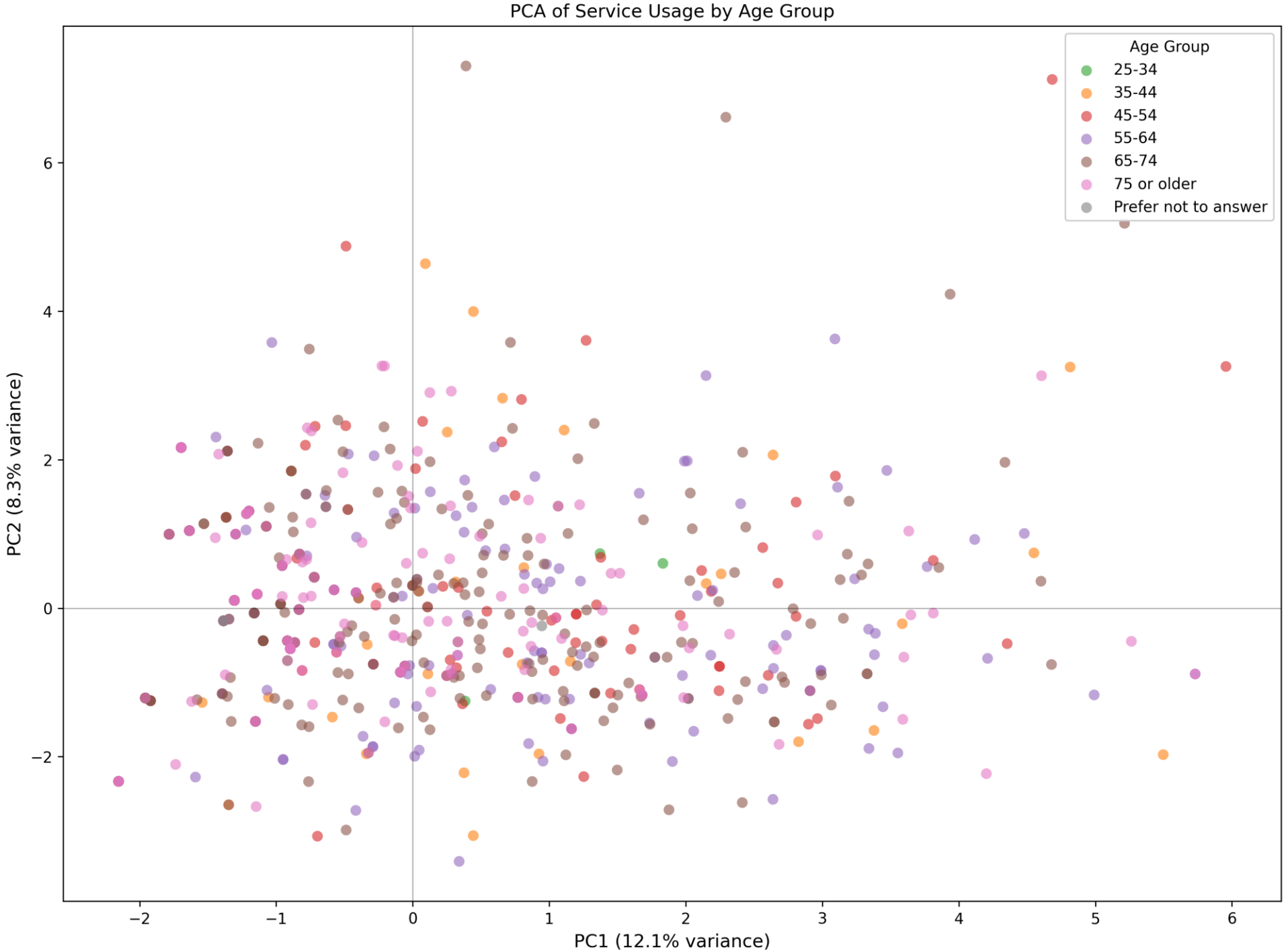
Finding client patterns

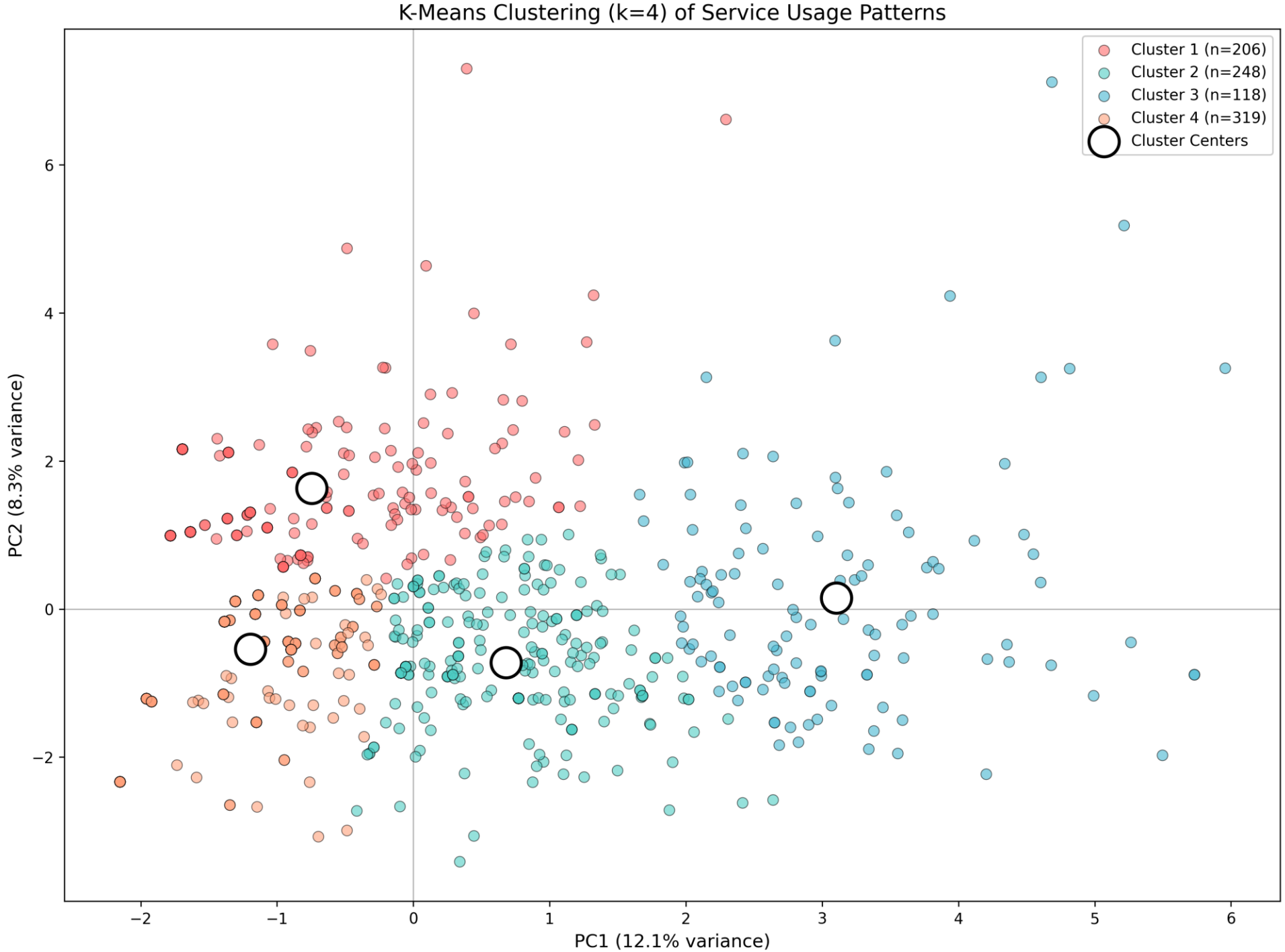
```
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,70
0,1,0,0,0,0,0,0,0,0,1,0,1,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,60
0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,1,0,0,0,0,0,0,0,1,0,0,1,0,70
0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,70
1,0,0,0,0,0,1,0,0,0,1,0,0,0,1,1,0,0,0,0,0,0,1,0,0,0,0,0,60
1,1,0,0,1,1,0,0,0,0,1,1,1,1,0,1,0,0,0,0,0,0,0,0,0,0,0,0,60
0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,1,0,70
0,0,1,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,80
1,0,0,0,0,0,0,0,0,0,1,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,80
1,0,0,1,0,0,0,1,0,0,1,0,0,0,0,1,0,1,1,0,0,0,0,0,0,0,0,60
```

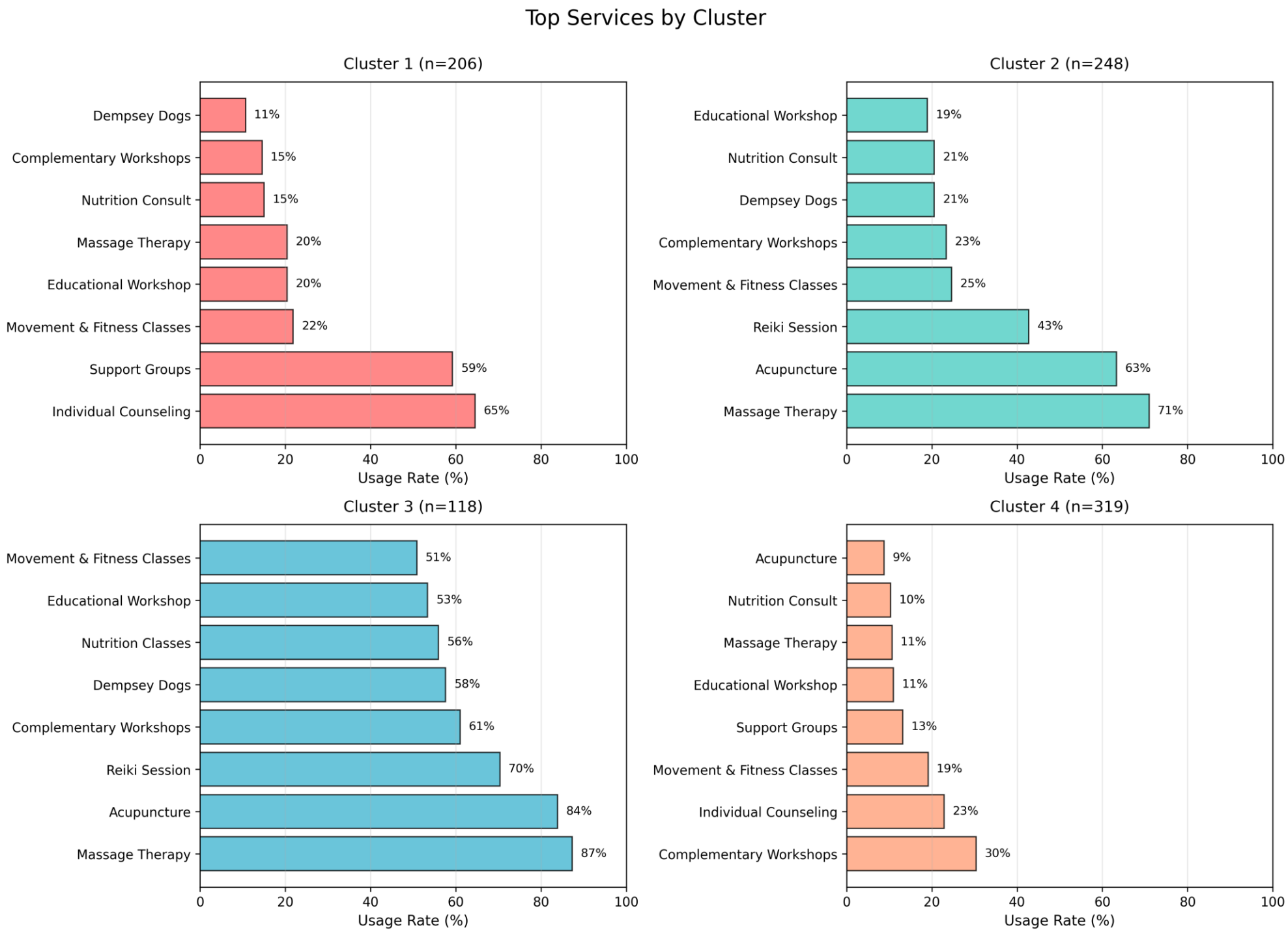

Age Group Prediction Model Accuracies
Accuracy (%)
100
80
60
40
20
0
38.37%
12.79%
26.74%
20.93%
NaiveBayes
GaussianNB
KNN (k=3)
DecisionTree
Model

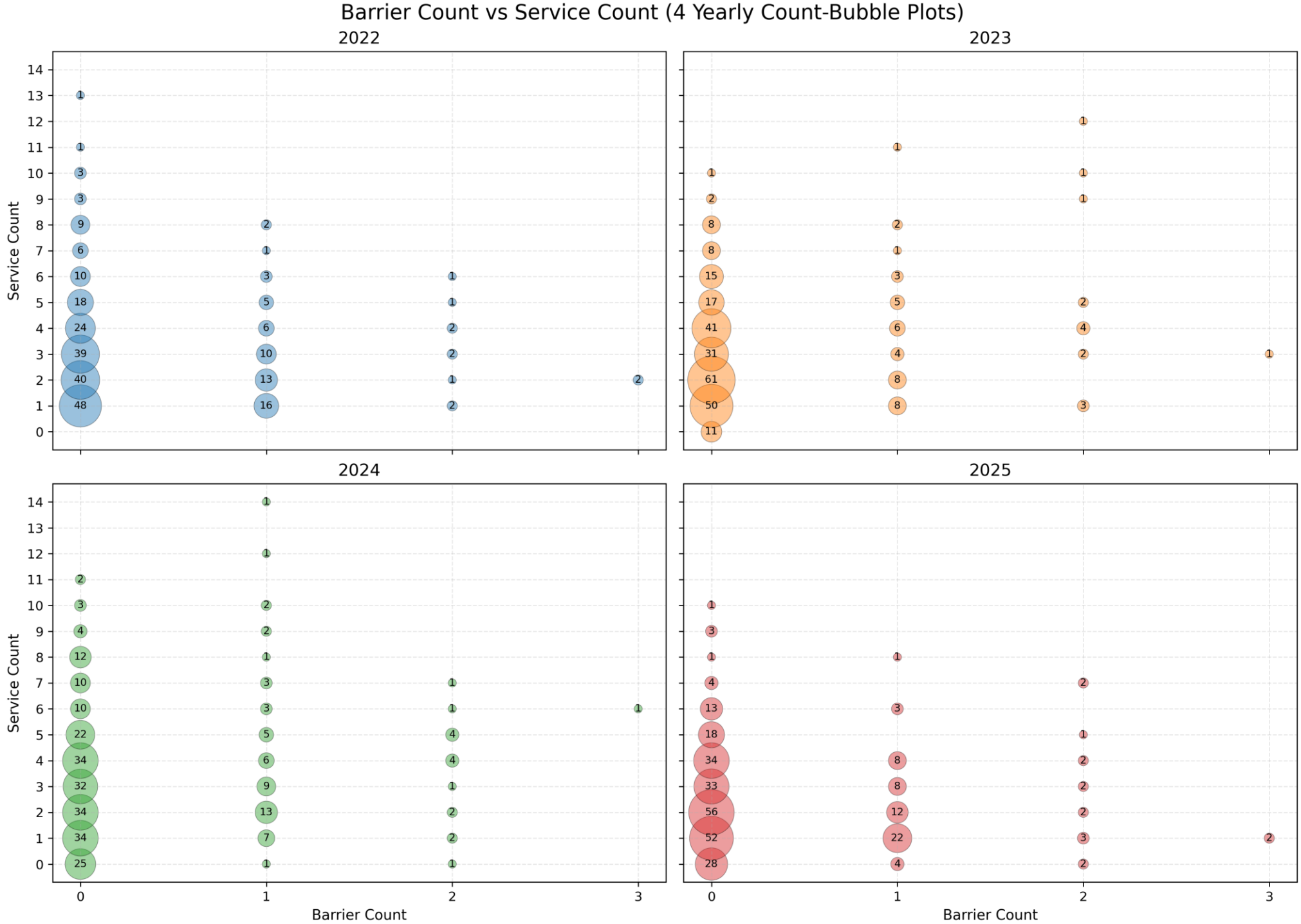

PC1 loadings
Massage Therapy
Acupuncture
Reiki Session
Dempsey Dogs
Nutrition Classes
Educational Workshop
Dempsey Soup Program
Nutrition Consult
Movement & Fitness Consult
Comfort Items
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Loading


PC2 loadings
Individual Counseling
Support Groups
Expressive Arts Workshop
Youth & Family Classes
Youth & Family Counseling
Acupuncture
Massage Therapy
Legacy & Life Reflections
−0.4
−0.2
0.0
0.2
0.4
Loading


PCA of Service Usage by Age Group
Age Group
25-34
35-44
45-54
55-64
65-74
75 or older
Prefer not to answer
PC2 (8.3% variance)
PC1 (12.1% variance)
6
4
2
0
−2
−2
−1
0
1
2
3
4
5
6


K-Means Clustering (k=4) of Service Usage Patterns
Cluster 1 (n=206)
Cluster 2 (n=248)
Cluster 3 (n=118)
Cluster 4 (n=319)
Cluster Centers
PC2 (8.3% variance)
PC1 (12.1% variance)
6
4
2
0
−2
−2
−1
0
1
2
3
4
5
6


Top Services by Cluster
Cluster 1 (n=206)
Dempsey Dogs 11%
Complementary Workshops 15%
Nutrition Consult 15%
Massage Therapy 20%
Educational Workshop 20%
Movement & Fitness Classes 22%
Support Groups 59%
Individual Counseling 65%
0 20 40 60 80 100
Usage Rate (%)
Cluster 2 (n=248)
Educational Workshop 19%
Nutrition Consult 21%
Dempsey Dogs 21%
Complementary Workshops 23%
Movement & Fitness Classes 25%
Reiki Session 43%
Acupuncture 63%
Massage Therapy 71%
0 20 40 60 80 100
Usage Rate (%)
Cluster 3 (n=118)
Movement & Fitness Classes 51%
Educational Workshop 53%
Nutrition Classes 56%
Dempsey Dogs 58%
Complementary Workshops 61%
Reiki Session 70%
Acupuncture 84%
Massage Therapy 87%
0 20 40 60 80 100
Usage Rate (%)
Cluster 4 (n=319)
Acupuncture 9%
Nutrition Consult 10%
Massage Therapy 11%
Educational Workshop 11%
Support Groups 13%
Movement & Fitness Classes 19%
Individual Counseling 23%
Complementary Workshops 30%
0 20 40 60 80 100
Usage Rate (%)


Barrier Count vs Service Count (4 Yearly Count-Bubble Plots)
2022
2023
2024
2025
Service Count
Barrier Count

Results

- There are mainly two types of client: emotional and physical.
- Location and schedules are becoming a bigger problem over the years.
- The clients are getting older.
- Age is not a good indicator to predict their chosen services.
- Massage and counseling are always popular.

Issues

Issues/Challenges

- Data parsing is tricky. Services Choice and wording vary slightly over years.
- Client ratings are too high to get meaningful analysis.

Future Work

Future work

- In-depth analysis of qualitative data. Especially the negative comments on barriers.
- In depth analysis to build clearer client persona that does not rely on age.

Suggestions

Suggestions

- Capture the growing 65+ market.
- Expand service hours to reduce time barrier.
- Phase out low volume services.
- Gather critical feedback in future.

Thank you